Junyu Ren

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Research Interests

Verified agentic reasoning: eliminating agent hallucination with domain-specific formal verification.
Mathematical foundations for AI: complexity, topology, and mechanistic interpretability of neural networks.
AI4M, AI4S: efficient, cost-effective AI methods for non-incremental mathematical and scientific discovery.
Multi-model post-training: end-to-end RL for multi-agent systems with agent-specific independent models.

Education

University of Chicago 2024 – Present
PhD in Computational and Applied Mathematics.
Advisor: Lek-Heng Lim. GPA: 4.0/4.0.

University of Oxford 2023 – 2024
MMathPhil, Master of Mathematics and Philosophy. Distinction.
Thesis: *Koopman Operator Theory and Network-informed Dynamic Mode Decomposition.*
Supervisor: Renaud Lambiotte.

University of Oxford 2020 – 2023
BA, Mathematics and Philosophy. First Class; cohort ranking 1/20.

Research Experience

MATS Fellow (MATS 11) Sept. 2026 – present
Mentors: Sabrina Shih and Saad Siddiqui, Safe AI Forum (SAIF).
Agentic incident confidential verification using formal verification and zero-knowledge proofs.

Researcher Jun. 2026 – Oct. 2026
Scaling laws in the number of heterogeneous, specialized models post-trained within a model population.

Research Assistant, University of Chicago Apr. 2026 – present
Supervisor: Risi Kondor; collaborator: Su Hyeong Lee.
Mesa-optimization by rollout-awareness as a risk in recursive self-improvement.
Funding: UK AI Security Institute (AISI).

Research Assistant, University of Chicago Jul. 2025 – present
Supervisor: Lek-Heng Lim.
Neural network theory and interpretability using topology and geometry.

Research and Publications

[*] *Equal contribution / co-first authorship; co-first authors listed in alphabetical order.*

Published and Accepted

- **Manifold-Weighted Neural Networks.**
  Junyu Ren, Lek-Heng Lim.
  NeurIPS 2026 (accepted).
- **Optimization Dynamics Imprint Semantic Specificity in Contrastive Embedding Norms.**
  Ziwei Su, **Junyu Ren**, Victor Veitch.
  NeurIPS 2026 (accepted).
- **Low-dimensional Topology of Deep Neural Networks.**
  Junyu Ren, Lek-Heng Lim.
  ICML 2026.
- **EG-VAR: A Path Toward Eliminating LLM Hallucination in Empirical Inference via Tool-Attested Kernel Proofs.**
  Junyu Ren.
  Accepted at three ICML 2026 workshops — three parts of one agenda:
  FAGEN — proof-checker feedback for agentic workflows.
  TAIGR — formal sidecars as technical governance for high-stakes outputs.
  PhilML ("From Prompts to Proof Obligations") — world-directed reasoning as auditable formal commitments.

- **When Do Transformer Components Compose? Validating a Log-Pool Decomposition Criterion.**
  Junyu Ren, Su Hyeong Lee, Risi Kondor.
  ICML 2026 Workshop PhilML.
- **Routing-Mediated Structural Pseudo-Alignment in Sparse Mixture-of-Experts.**
  Junyu Ren, Su Hyeong Lee, Risi Kondor.
  ICML 2026 Workshop Mech Interp.
- **MAPPA: Scaling Multiagent Systems with Process Rewards.**
  Ed Li*, **Junyu Ren***, Cat Yan.
  ICLR 2026 Workshop RSI.
- **Agent-as-a-Coach: Towards Fully Agentic, Stateful, and Tool-Augmented Process Rewards.**
  Ed Li*, **Junyu Ren***, Cat Yan, Kerem Goksel.
  ICLR 2026 Workshop MALGAI.
- **Time-aware Feature Selection: Adaptive Temporal Masking for Stable Sparse Autoencoder Training.**
  Ed Li*, **Junyu Ren***.
  ICLR 2025 Workshop XAI4Science.
- **Unlocking Hierarchical Concept Discovery in Language Models Through Geometric Regularization.**
  Ed Li*, **Junyu Ren***.
  ICLR 2025 Workshop BuildingTrust.

Preprints

- **Pierce–Birkhoff conjecture is false.**
  Zehua Lai, Lek-Heng Lim, **Junyu Ren**.
  arXiv:2609.10420, Sep. 2026.
- **Build Your Personalized Research Group: A Multiagent Framework for Continual and Interactive Science Automation.**
  Ed Li*, **Junyu Ren***, Xintian Pan, Cat Yan, Chuanhao Li, Dirk Bergemann, Zhuoran Yang.
  arXiv:2510.15624, Oct. 2025.

Under Review and In Preparation

- **Training a 2-token attention is NP-hard.**
  Junyu Ren, Zehua Lai, Lek-Heng Lim.
  Manuscript under review, 2026.
- **Pierce–Birkhoff conjecture in dimension 3.**
  Junyu Ren, Zehua Lai, Lek-Heng Lim.
  Manuscript in preparation, 2026.
- **Hodge-decomposed Hochschild cohomology of socle quotient rings.**
  Junyu Ren, Lek-Heng Lim, Ke Ye.
  Manuscript in preparation, 2026.

Awards and Honors

McCormick Fellowship, University of Chicago 2024
Gibbs Prize for FHS Mathematics and Philosophy, University of Oxford 2023
Oriel College Scholarship, University of Oxford 2020 – 2024

Talks & Presentations

- **Low-dimensional topology of deep neural networks.**
  AMS 2026 Spring Eastern Sectional Meeting, Special Session, Boston College, Mar. 2026.
- **Hallucination as a Type Error.**
  IDEAL Workshop on Language Models, University of Illinois Chicago, May 2026.

Professional Services

Reviewer, ICLR 2027 2027
Reviewer, ICML 2026 *(Gold Reviewer Reward)* 2026
Reviewer, ICML 2026 Workshops 2026
Referee, Journal of the ACM (JACM) Aug. 2026 – present
Referee, Communications Physics (Nature Portfolio) 2026
Referee, The Journal of Geometric Analysis Jun. 2026 – present
Referee, Chaos, Solitons & Fractals Jun. 2026 – present
Referee, Linear Algebra and Its Applications May 2025 – present
Referee, SIAM Journal on Computing Oct. 2025 – present
Referee, SIAM Journal on Matrix Analysis and Applications Oct. 2025 – present
Teaching Assistant, University of Chicago Oct. 2025 – present